

Technical and Business Writing

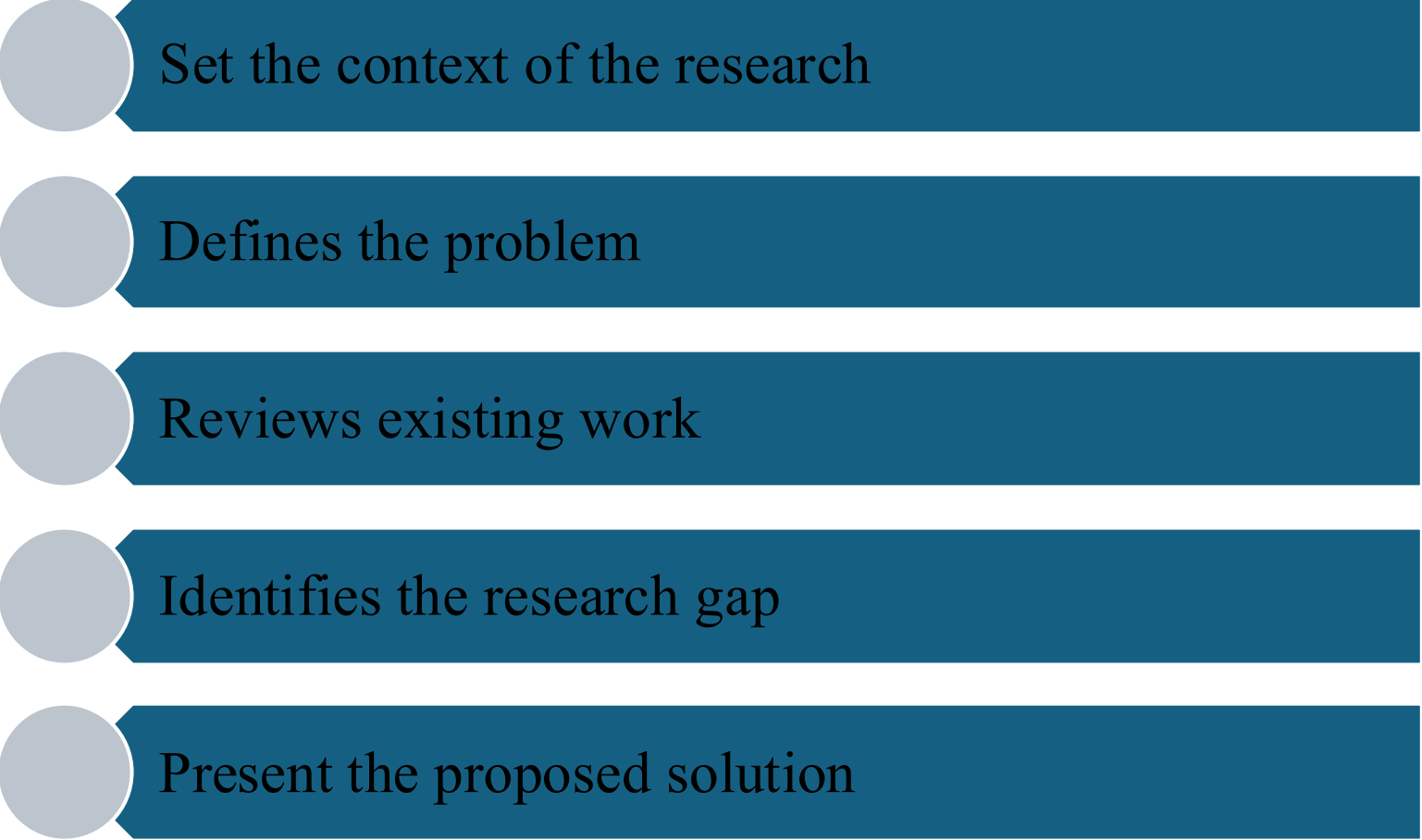
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How to write the introduction of a technical research paper

What is an Introduction?



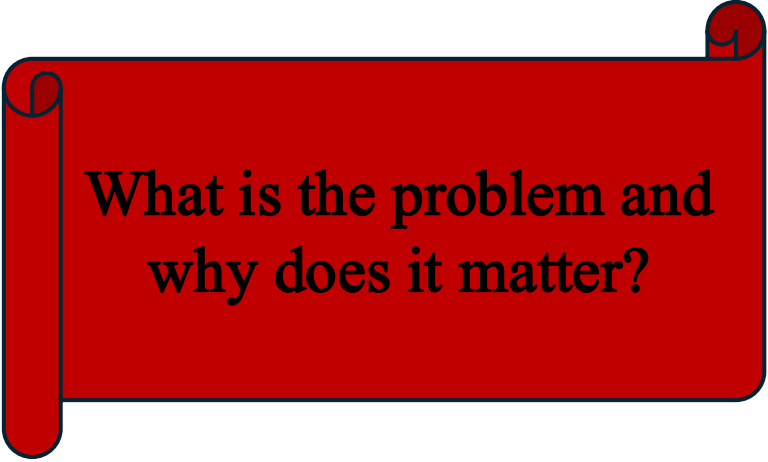
Set the context of the research

Defines the problem

Reviews existing work

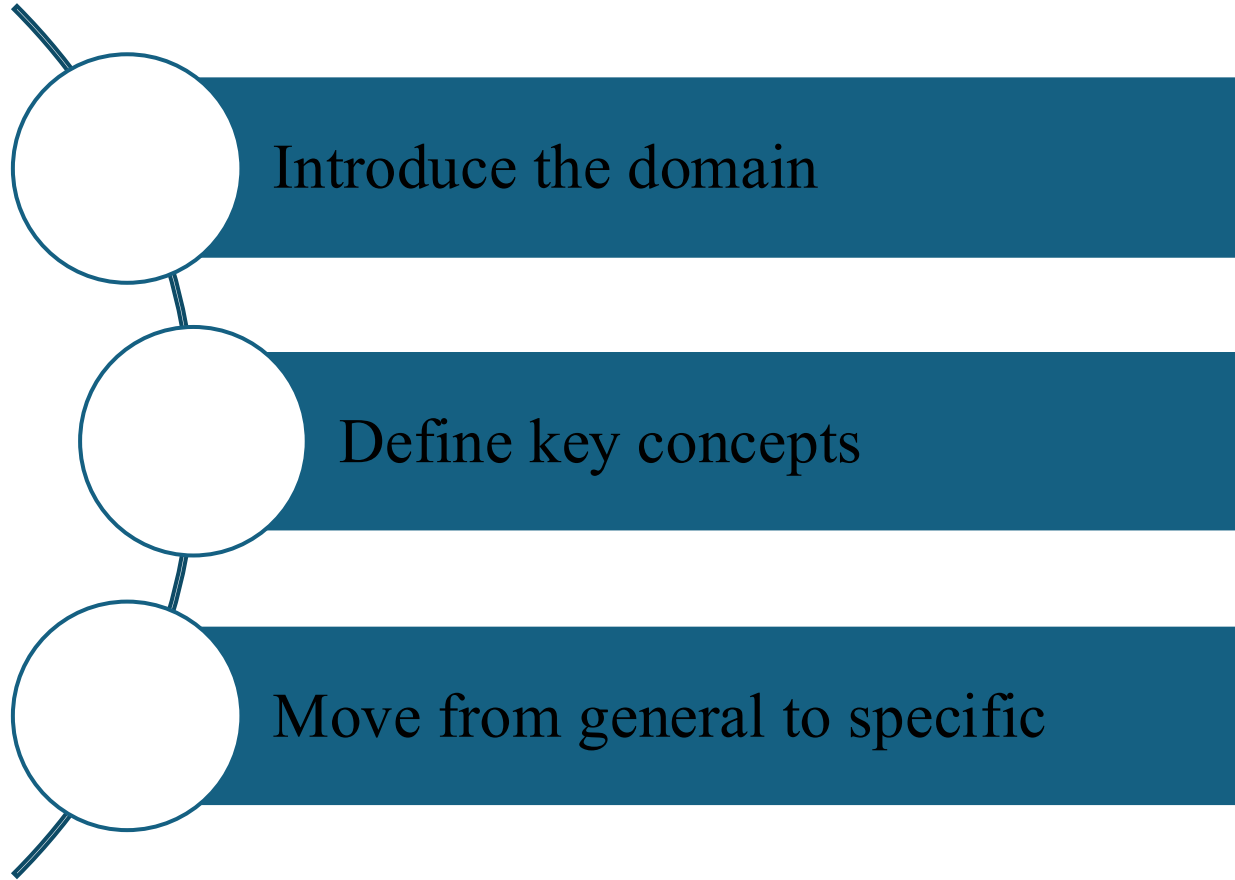
Identifies the research gap

Present the proposed solution



What is the problem and
why does it matter?

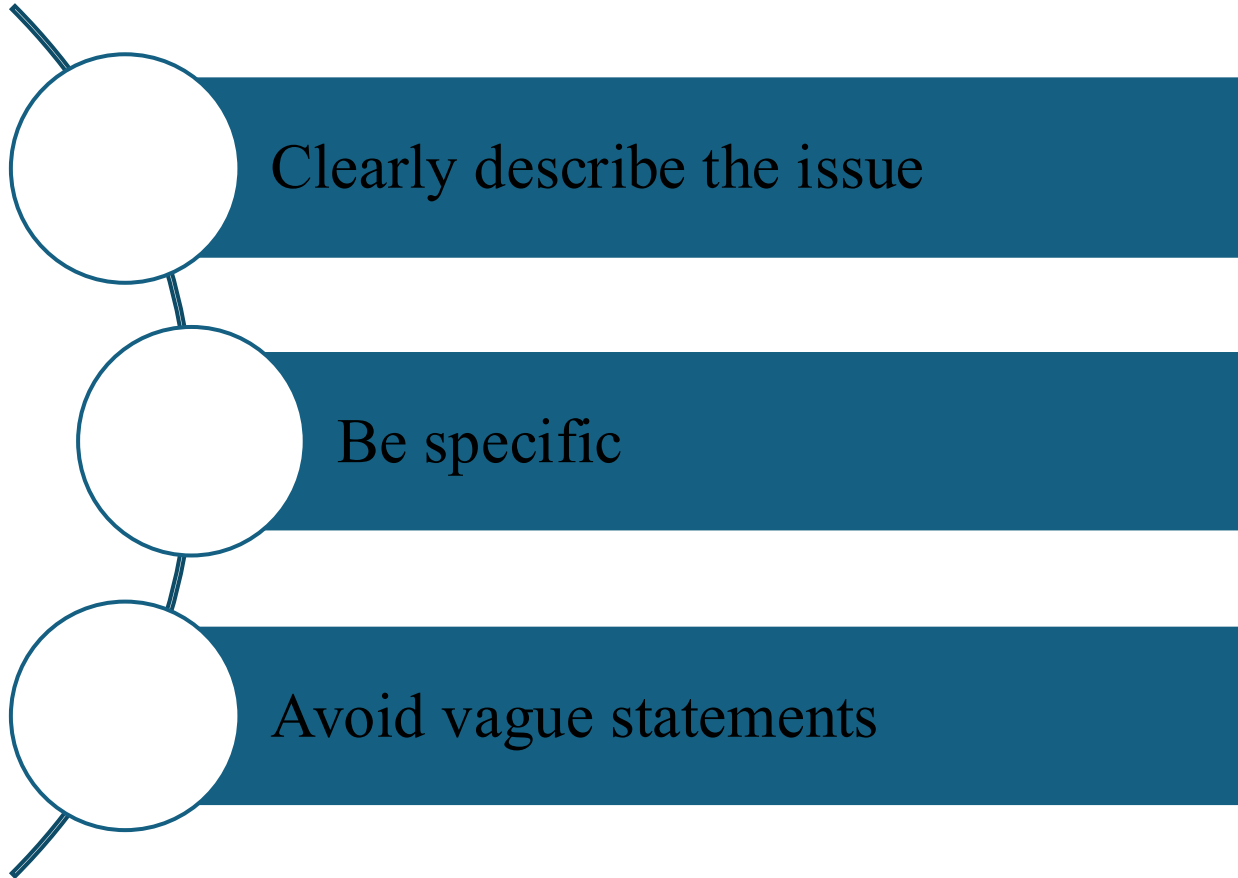
Structure of a strong introduction: Background context



Example

- Explain retina and its function
- Its role in diagnosis

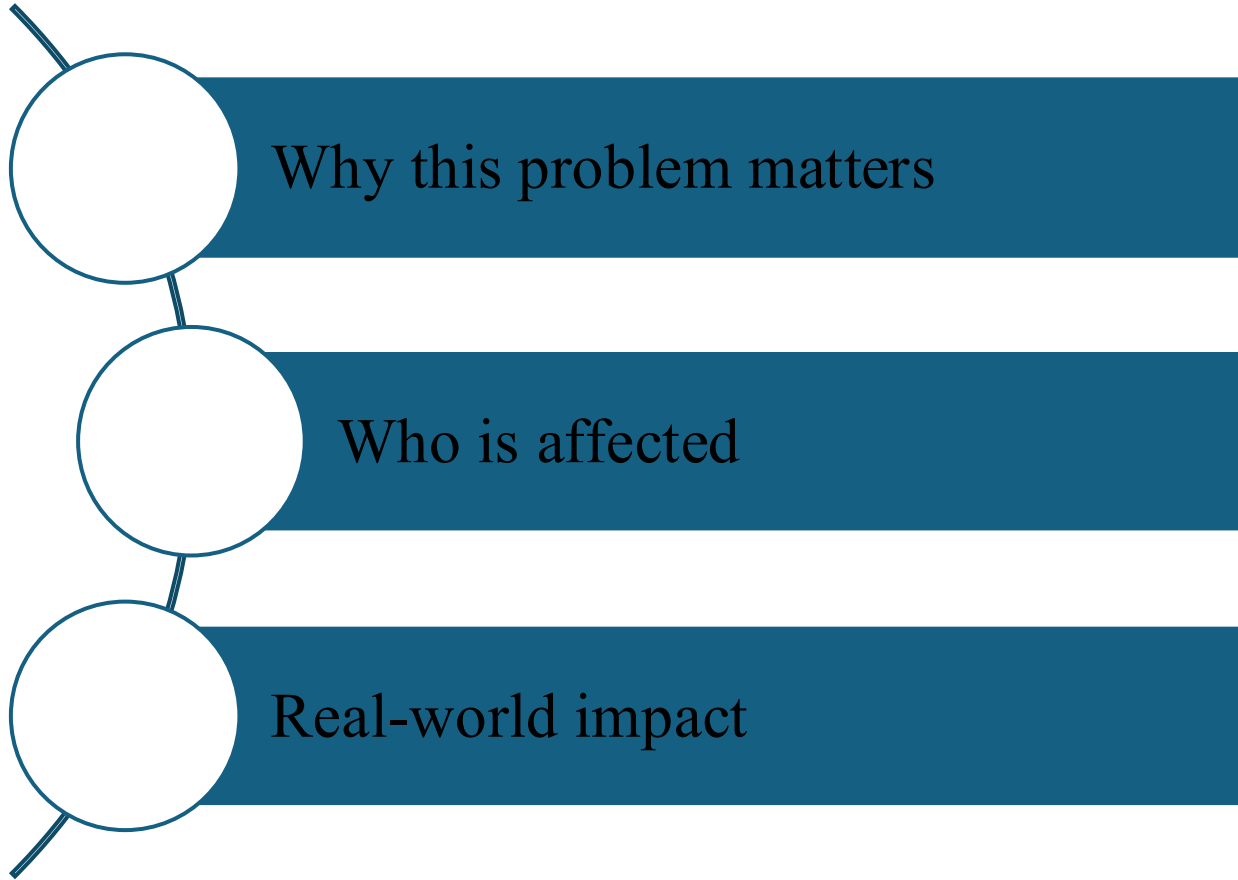
Structure of a strong introduction: Problem



Example

- Retinal diseases cause blindness
- Early detection is difficult

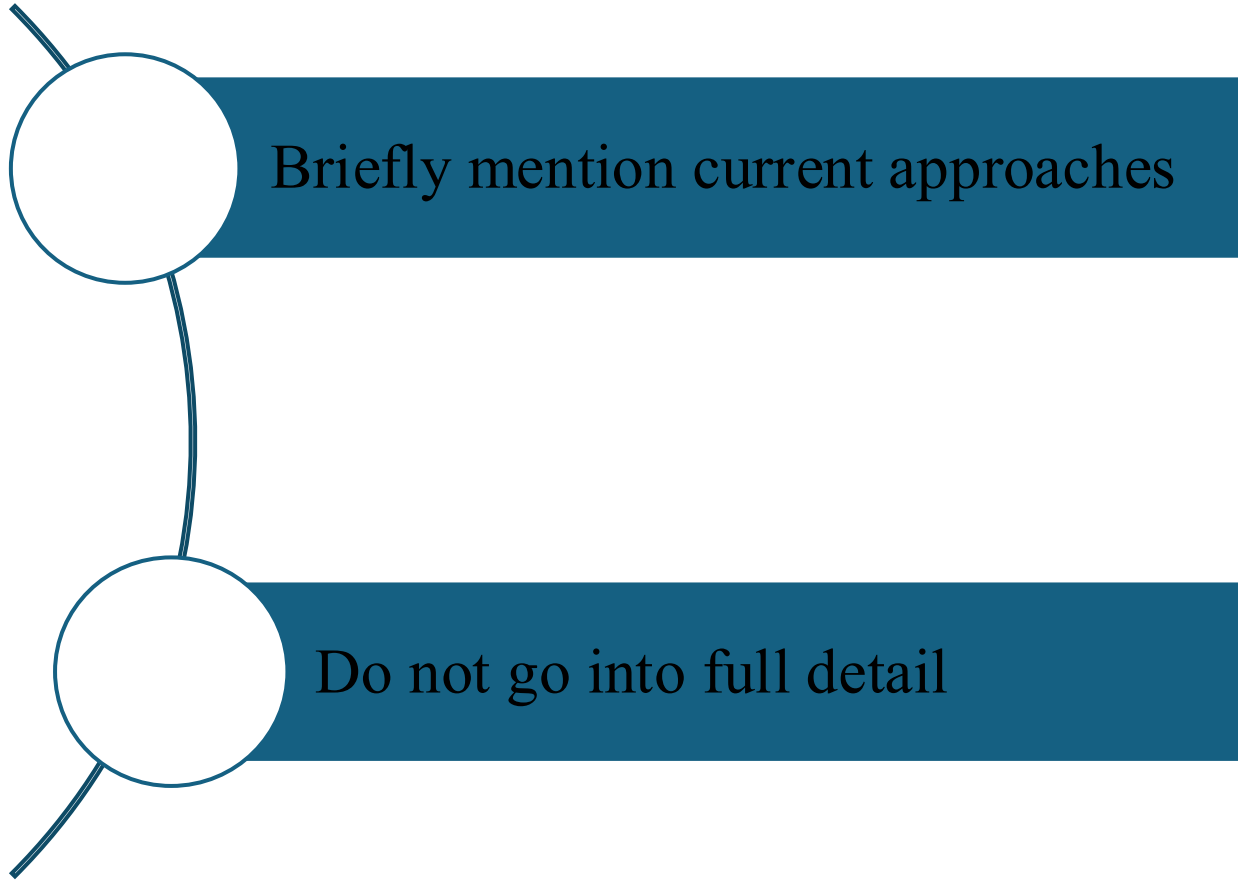
Structure of a strong introduction: Importance of Problem



Example

- Millions affected globally
- Early diagnosis prevents vision loss

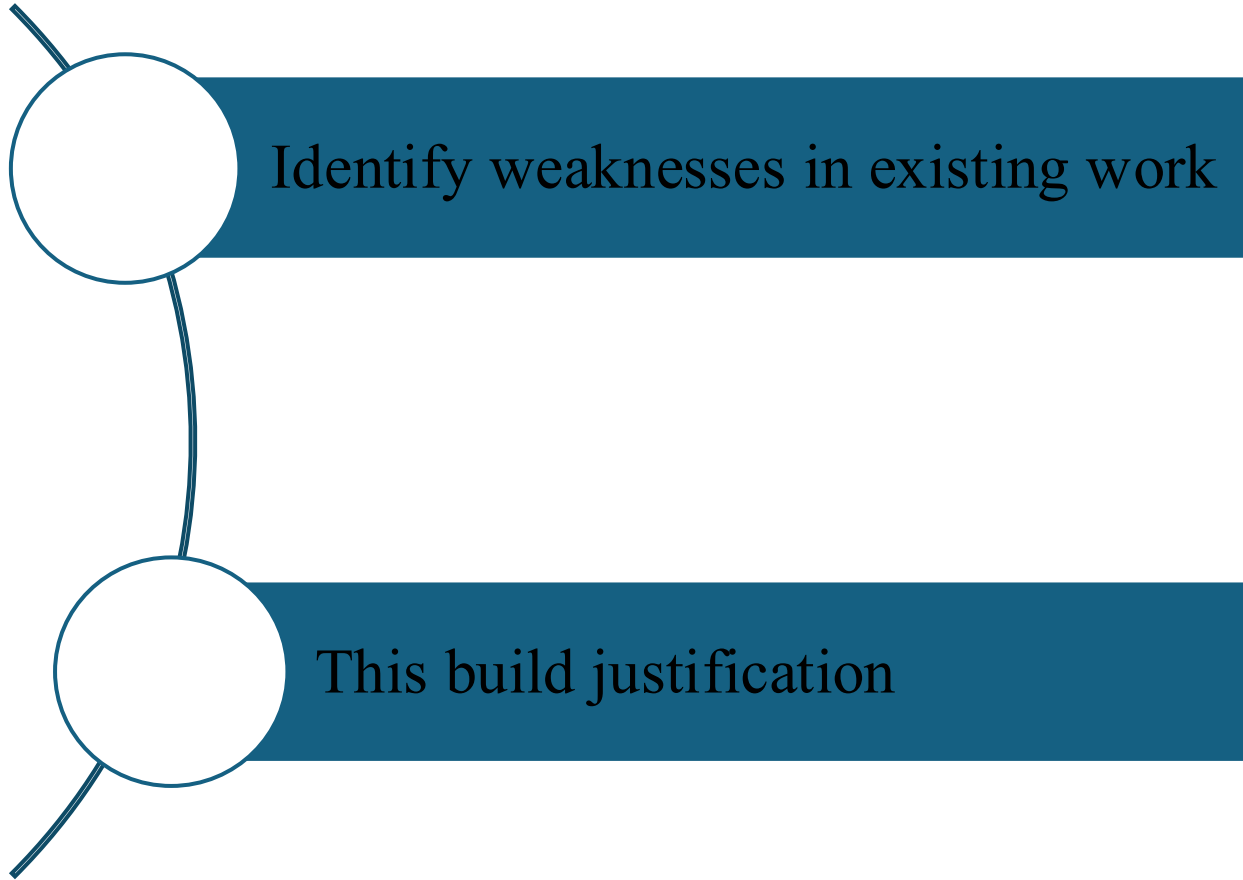
Structure of a strong introduction: Existing solution (mini literature)



Example

- CAD systems
- Machine learning methods
- Deep learning approached

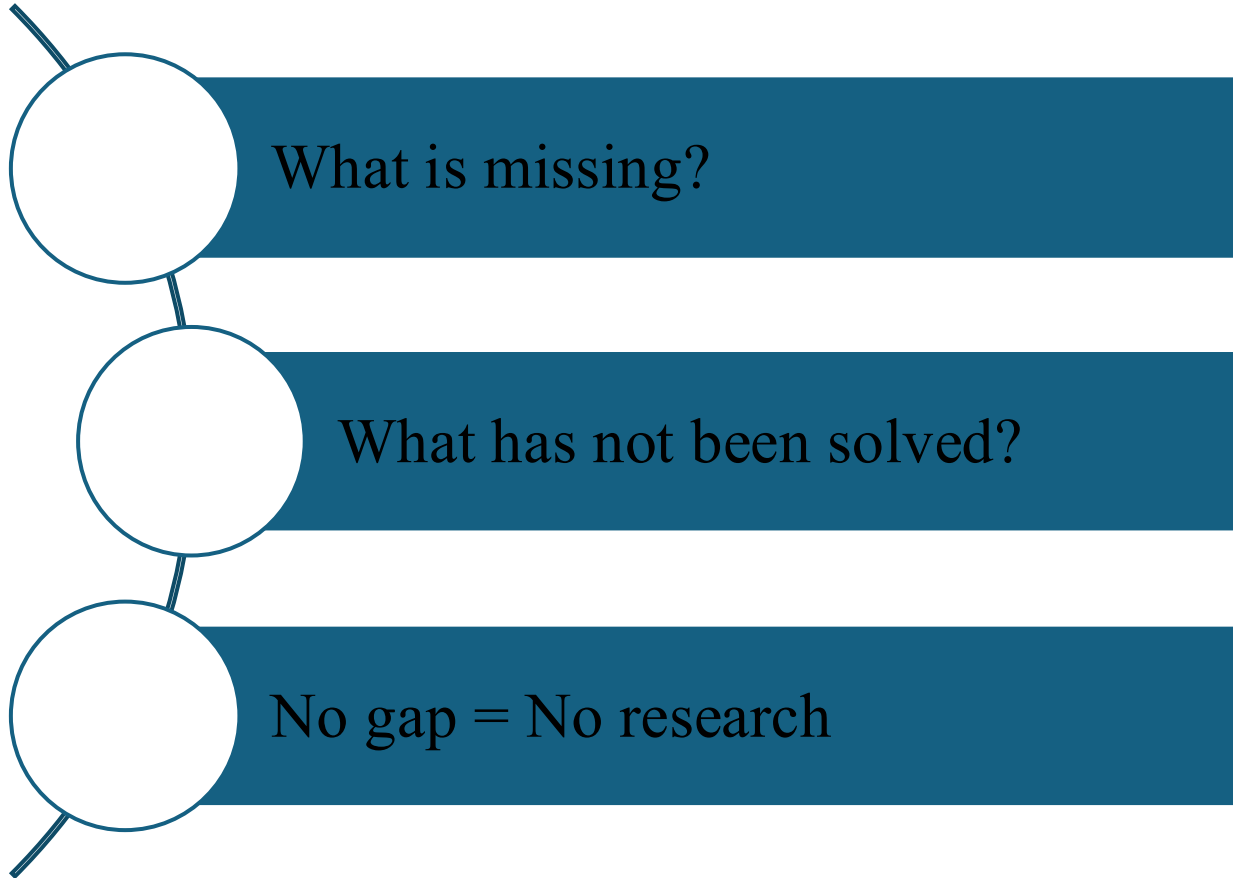
Structure of a strong introduction: Limitations



Example

- Not accurate
- Focus on single disease

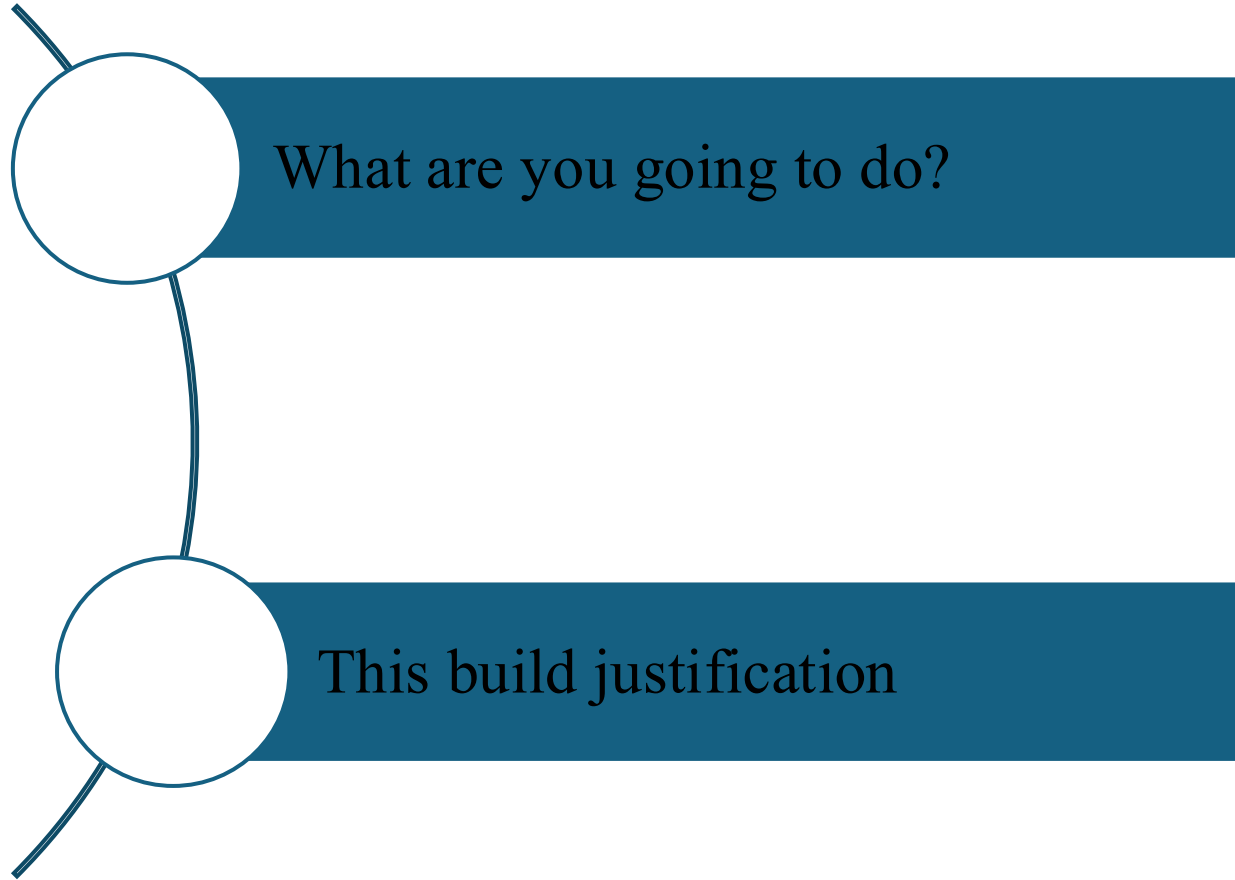
Structure of a strong introduction: Research Gap



Example

- Existing models cannot detect multiple diseases simultaneously

Structure of a strong introduction: Proposed Solution



Example

- Develop multi-label classification system
- Use transformer-based model

Where do we need citation in the introduction?

When you state statistics

For example: Diseases such as ARMD and glaucoma cause blindness to millions of people...

This must be cited because:

- It is factual data
- Not original idea by the author

When you refer to existing research

For example: Deep learning models such as CNNs have been used for disease detection.

Cite because:

- Refers to previous studies

Where do we need citation in the introduction?

When you mention methods/technologies used by others

For example: CAD systems and machine learning techniques have been applied...

Cite because:

- These approaches come from prior work.

When you make specific claims based on research

For example: CNN-based models achieve high accuracy in image classification.

Needs citation because:

- It is a research-backed claim

Where you do not need citation?

Your observation in the problem statement

For example: Drivers spend time searching for parking...

No citation because:

- This can be your observation

Your research gap (usually no citation)

For example: Existing systems do not support multi-label classification.

No citation because:

General/common Knowledge

- For example: The retina is part of the human eye

Ask yourself these three questions:

- ✓ Did I get this from a source? →  Cite
- ✓ Is this a research claim or data? →  Cite
- ✓ Is this my idea or interpretation? →  No citation

Task(s) for attendance

- Task 1: Identify elements of introduction from sample introduction section of a technical research paper
- Task 2: Expand your problem statement into introduction section